Answer Key & Teacher Notes

Session 1 — Customary vs. Metric Conversions · Session 2 — Multi-Step System Conversions and Dosage

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Customary vs. Metric Conversions

TALK PROMPTS

- Why is checking reasonableness so important when converting between systems?
- A meter is bigger than a foot. Why should the number of feet come out larger than the number of meters?

PARTNER TASK

Soccer Field, Suitcase, Backpack Check. One partner predicts whether the new number will be bigger or smaller; the other converts with the given fact. Compare the result with the prediction, then switch roles.

WATCH FOR

- Dividing when the new unit is smaller (50 mi ÷ 1.6 = 31.25 km).
- Comparing amounts in different units, such as 70 m and 160 ft, without converting.
- Treating an answer from a rounded fact (≈) as exact.

SESSION 2 · TEACHER NOTES

Multi-Step System Conversions and Dosage

TALK PROMPTS

- How can labeling units help you choose the correct operation in every step?
- What would go wrong if you skipped the first step and multiplied the pony's pounds by 2?

PARTNER TASK

Dog Medicine, Elephant Calf, Gas Tank. One partner does step 1 and writes its unit; the other checks that the unit matches the rate before doing step 2. Switch roles for each problem.

WATCH FOR

- Using pounds with a per-kilogram rate (66 lb × 4 mg instead of 30 kg × 4 mg).
- Multiplying by 2.2 to change pounds to kilograms.
- Adding length and width instead of multiplying to find the garden area (15 instead of 54 ft²).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.3d	1	2	2	2	1	1	2	3	2	1	1
S1 · Form B	6.RP.A.3d	1	2	2	2	1	1	2	3	2	1	1
S2 · Form A	6.RP.A.3d	1	2	2	2	1	2	2	3	2	1	1
S2 · Form B	6.RP.A.3d	1	2	2	2	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

conversion factor: A fact that **links** two units.

approximately: About. The symbol ≈ means "about **equal** to."

reasonableness: Checking that your answer's **size** makes sense.

- Use the fact the problem gives. Many facts are rounded, so the answer is an **estimate**.
- Changing to a smaller unit, **multiply**. Changing to a bigger unit, **divide**.

Example: The fact: 1 kg ≈ **2.2** lb. A kilogram is heavier than a pound, so the number gets smaller: **divide**. 44 ÷ 2.2 = 20. Check: 20 × 2.2 = 44. Answer: 44 lb ≈ **20** kg.

YOU TRY

1. a. 6.6 lb b. 25.4 cm

2. 5 ÷ 1.6 = 3.125, about 3.1 miles

EXIT TICKET

1. a. 44 lb b. 10 mi

2. 6 × 3.28 = 19.68 ft

3. **A** (multiply by 3.28)

PRACTICE FORM A

1. a. 22 lb b. 20 kg c. 16.4 ft d. 10.16 cm

2. a. 100 ÷ 1.6 = 62.5 mph b. Less, because miles are longer than kilometers, so it takes fewer of them.

3. a. 70 × 3.28 = 229.6 ft b. The soccer field

4. a. 20 × 2.2 = 44 lb b. Yes, under by 6 lb

5. 150 × 1.6 = 240 km

6. 16.9 × 29.6 = 500.24, about 500 mL

7. 7 × 2.2 = 15.4 lb; yes, it is under 16 lb

8. a. 4 × 3.28 = 13.12 ft b. The rope, by about 1.12 ft

9. A kilometer is shorter than a mile, so there must be MORE kilometers than miles. Multiply: 50 × 1.6 = 80 km.

10. **B** (6.56)

11. **D** (2)

Self-check word **METRIC**

PRACTICE FORM B

1. a. 33 lb b. 30 kg c. 13.12 ft d. 7.62 cm

2. a. 80 ÷ 1.6 = 50 mph b. Less, because miles are longer than kilometers, so it takes fewer of them.

3. a. 65 × 3.28 = 213.2 ft b. The soccer field

4. a. 24 × 2.2 = 52.8 lb b. No, over by 2.8 lb

5. 120 × 1.6 = 192 km

6. 12 × 29.6 = 355.2, about 355 mL

7. 6 × 2.2 = 13.2 lb; no, it is over 12 lb

8. a. 3 × 3.28 = 9.84 ft b. The cable, by about 0.66 ft

9. A kilometer is shorter than a mile, so there must be MORE kilometers than miles. Multiply: 40 × 1.6 = 64 km.

10. **C** (9.84)

11. **A** (3)

Self-check word **LENGTH**

SESSION 2 · ANSWERS

Multi-Step System Conversions and Dosage

GUIDED NOTES

multi-step: Do two or more operations in **order**.

dose: The **amount** of medicine to give.

per: For **each** one.

- First change the given amount to the unit the **rate** uses.
- Write the **unit** after every step so you know which operation comes next.

Example: Pounds to kilograms: 11 ÷ 2.2 = **5** kg. Dose: 5 kg × 3 mg per kg = **15** mg. Syrup: 15 mg ÷ 5 mg per mL = **3** mL. Answer: The cat gets **3** mL of syrup.

YOU TRY

1. a. 20 kg b. 60 mg

2. 4 × 6 = 24 ft²; 24 ÷ 8 = 3 bags

EXIT TICKET

1. a. 25 kg b. 50 mg

2. 180 ÷ 12 = 15 L; 15 × $2 = $30

3. **A** (before you use the rate)

PRACTICE FORM A

1. a. 15 kg b. 75 mg c. 5 mL d. 55 lb

2. a. 187 ÷ 2.2 = 85 kg b. 85 × 2 = 170 mg

3. a. 110 × 2.2 = 242 lb b. 2,000 − 242 = 1,758 lb

4. a. 57 ÷ 3.8 = 15 gal b. 15 × $3.50 = $52.50

5. 280 ÷ 28 = 10 oz

6. 66 ÷ 2.2 = 30 kg; 30 × 4 = 120 mg

7. a. 240 ÷ 12 = 20 L b. 20 × $1.80 = $36.00

8. a. 6 × 9 = 54 ft² b. 54 ÷ 6 = 9 bags c. 9 × $4.25 = $38.25 d. Only 54 ÷ 9 = 6 bags are needed, so the cost drops to $25.50.

9. A kilogram is heavier than a pound, so the number of kilograms must be smaller than 44. Divide: 44 ÷ 2.2 = 20 kg.

10. **D** (20)

11. **B** (60 mg)

Self-check word **DOSAGES**

PRACTICE FORM B

1. a. 10 kg b. 60 mg c. 5 mL d. 66 lb

2. a. 176 ÷ 2.2 = 80 kg b. 80 × 3 = 240 mg

3. a. 120 × 2.2 = 264 lb b. 2,000 − 264 = 1,736 lb

4. a. 76 ÷ 3.8 = 20 gal b. 20 × $3.80 = $76.00

5. 196 ÷ 28 = 7 oz

6. 88 ÷ 2.2 = 40 kg; 40 × 5 = 200 mg

7. a. 300 ÷ 12 = 25 L b. 25 × $1.60 = $40.00

8. a. 8 × 9 = 72 ft² b. 72 ÷ 8 = 9 bags c. 9 × $3.75 = $33.75 d. Only 72 ÷ 12 = 6 bags are needed, so the cost drops to $22.50.

9. A kilogram is heavier than a pound, so the number of kilograms must be smaller than 55. Divide: 55 ÷ 2.2 = 25 kg.

10. **B** (30)

11. **A** (60 mg)

Self-check word **WEIGHTS**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · CUSTOMARY VS. METRIC CONVERSIONS

1. a. 15.24 cm b. 40 km c. 5 kg d. 11.4 L e. 110 lb f. 5 mi g. 2 kg h. 16 km

2. 5 kg ≈ 11 lb; 22 lb ≈ 10 kg; 25 kg ≈ 55 lb

3. 96 ÷ 1.6 = 60; about 60 miles

4. a. 55 ÷ 2.2 = 25; about 25 kg b. No. 25 kg > 23 kg, so it is about 2 kg over the limit

5. 100 m ≈ 100 × 1.09 = 109 yd, which is more than 100 yd. The 100-meter race is longer (by about 9 yards).

SHEET 2 · MULTI-STEP SYSTEM CONVERSIONS AND DOSAGE

1. a. 35 kg b. 140 mg c. 6 tsp d. 1,500 mL

2. a. 8.8 ÷ 2.2 = 4 kg b. 4 × 5 = 20 mg

3. a. 4 × 400 × 5 = 8,000 m b. 8 km c. 8 ÷ 1.6 = 5 miles

4. a. 2 × 5 = 10; about 10 mL b. 120 ÷ 10 = 12; about 12 doses

5. She used pounds, but the rate is per kilogram. 66 ÷ 2.2 = 30 kg; 30 × 3 = 90 mg.

6. **A** (30 mg)

7. **A** (2.85 L)

October 2026 conceptual additions — teacher key

Session 1 · Customary vs. Metric Conversions

15.4 ÷ 2.2 = 7, so 15.4 lb ≈ 7 kg. A kilogram corresponds to more than one pound, so the numerical measure decreases. Check 7 × 2.2 = 15.4 using the approximate factor.

Extension: The factor 2.2 is itself approximate. Exact arithmetic with a rounded factor still produces an approximate conversion.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Multi-Step System Conversions and Dosage

11 lb ≈ 5 kg; 5 × 3 = 15 g pigment; 15 ÷ 5 = 3 scoops. The unit path is pounds → kilograms → grams pigment → scoops. The result is approximate because the conversion factor is approximate.

Extension: An early rounded intermediate value changes the input to later operations; keep the available precision until the requested final rounding.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.